Problem Set 4D

Vector projections

Do these after the slides for Section 4D. Textbook Exercise 4D: Questions 7, 9, 10, 11. Answers are on the last page. Show full working in your exercise book.

Skills

1 Find the scalar resolute of $\mathbf{a}$ in the direction of $\mathbf{b}$.

(a) $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k},\ \mathbf{b} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

(b) $\mathbf{a} = \mathbf{i} + 5\mathbf{j} - 3\mathbf{k},\ \mathbf{b} = 3\mathbf{i} - 4\mathbf{k}$

2 Find the vector resolute of $4\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ in the direction of $\mathbf{i} - \mathbf{j} + \mathbf{k}$.

3 Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$. Find:

(a) the scalar resolute of $\mathbf{a}$ in the direction of $\mathbf{b}$

(b) the vector resolute of $\mathbf{a}$ in the direction of $\mathbf{b}$

(c) the vector resolute of $\mathbf{a}$ perpendicular to $\mathbf{b}$

4 Find the distance from the point $A(2, 3, 6)$ to the line through the origin and $B(1, 2, 2)$.

5 The scalar resolute of $m\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ in the direction of $2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ is zero. Find m.

Exam style

6 The vector resolute of $\mathbf{a} = 2\mathbf{i} + m\mathbf{j} + \mathbf{k}$ in the direction of $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ is $2\mathbf{b}$. Find m.

7 For the points $A(1, 2, -1)$, $B(3, 3, 1)$ and $C(2, 5, 0)$, find the vector resolute of $\overrightarrow{AC}$ in the direction of $\overrightarrow{AB}$, and hence the distance from C to the line AB.

8 Write $\mathbf{a} = 5\mathbf{i} + 5\mathbf{j}$ as the sum of a vector parallel to $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and a vector perpendicular to $\mathbf{b}$.

Problem Set 4D: answers

1 (a) $\frac{8}{3}$ (b) 3

2 $\frac{1}{3}(\mathbf{i}-\mathbf{j}+\mathbf{k})$

3 (a) $-\frac{8}{3}$ (b) $-\frac{8}{9}(\mathbf{i}+2\mathbf{j}-2\mathbf{k})$ (c) $\frac{1}{9}(26\mathbf{i}+7\mathbf{j}+20\mathbf{k})$

4 $\dfrac{\sqrt{41}}{3}$

5 $m=-\frac{1}{2}$

6 $\dfrac{2m+4}{9}=2$, so $m=7$

7 $\frac{7}{9}(2\mathbf{i}+\mathbf{j}+2\mathbf{k})$; distance $=\sqrt{11-\frac{49}{9}}=\dfrac{5\sqrt{2}}{3}$

8 $\mathbf{a}=\frac{5}{3}(\mathbf{i}+2\mathbf{j}-2\mathbf{k})+\frac{5}{3}(2\mathbf{i}+\mathbf{j}+2\mathbf{k})$